1. The Transport Layer (Layer 4) in the OSI/TCP-IP reference model is fundamentally responsible for:

(A) End-to-end, process-to-process communication between application programs
(B) Node-to-node frame delivery across a single link
(C) Host-to-host packet routing across the Internet
(D) Physical transmission of electrical signals

2. Processes communicating across a network are uniquely identified at the Transport Layer using:

(A) Port Numbers (16-bit integers)
(B) MAC Addresses (48-bit hex)
(C) IP Addresses (32-bit dotted-decimal)
(D) Autonomous System Numbers

3. The combination of an IP address and a Port number is formally termed a:

(A) Socket Address (e.g., 192.168.1.5:80)
(B) Hardware Address
(C) URL Link
(D) Session Token

4. What is the total range of available 16-bit port numbers?

(A) 0 to 65,535 (2^16 ports)
(B) 0 to 1,023
(C) 0 to 32,767
(D) 0 to 255

5. Port numbers ranging from 0 to 1023 are officially designated by IANA as:

(A) Well-Known Ports (assigned to ubiquitous system services)
(B) Registered Ports
(C) Dynamic or Ephemeral Ports
(D) Private Experimental Ports

6. Which well-known port is standardly assigned to HTTP web server traffic?

(A) Port 80
(B) Port 443
(C) Port 21
(D) Port 25

7. Which well-known port is assigned to secure HTTPS encrypted web traffic?

(A) Port 443
(B) Port 80
(C) Port 8080
(D) Port 22

8. Which standard port is used by Domain Name System (DNS) queries over UDP?

(A) Port 53
(B) Port 67
(C) Port 110
(D) Port 143

9. Which port is standardly utilized for Secure Shell (SSH) remote administrative login?

(A) Port 22
(B) Port 23 (Telnet)
(C) Port 25 (SMTP)
(D) Port 53 (DNS)

10. Port numbers in the range 49,152 to 65,535 are designated as:

(A) Dynamic, Private, or Ephemeral Ports allocated temporarily to client programs
(B) Well-Known Ports
(C) Registered Service Ports
(D) Multicast Ports

11. The two principal Transport Layer protocols in the standard Internet Protocol Suite are:

(A) Transmission Control Protocol (TCP) and User Datagram Protocol (UDP)
(B) IP and ICMP
(C) HTTP and FTP
(D) Ethernet and Wi-Fi

12. User Datagram Protocol (UDP, RFC 768) is characterized as:

(A) A connectionless, lightweight, unreliable, best-effort transport service
(B) A connection-oriented reliable byte-stream protocol
(C) A full-duplex sliding window protocol with retransmission
(D) A hop-by-hop routed protocol

13. Transmission Control Protocol (TCP, RFC 793) is characterized as:

(A) A connection-oriented, reliable, full-duplex byte-stream delivery protocol
(B) A connectionless datagram delivery protocol
(C) An unacknowledged multicast streaming service
(D) A stateless frame multiplexer

14. The total fixed header length of a UDP packet (datagram) is:

(A) 8 Bytes (64 bits)
(B) 20 Bytes
(C) 40 Bytes
(D) 16 Bytes

15. The four fields comprising the 8-byte UDP header are Source Port, Destination Port, Checksum, and:

(A) Length (Length of header plus data in bytes)
(B) Sequence Number
(C) Acknowledgment Number
(D) Window Size

16. What happens when a UDP datagram encounters bit errors during transit and the checksum verification fails?

(A) The receiving host discards the corrupted datagram silently (or logs a warning)
(B) The receiver asks the sender to retransmit the lost packet
(C) The receiver repairs the bits automatically using FEC
(D) The sender pauses transmission

17. UDP is ideally suited for applications that prioritize:

(A) Low latency and real-time streaming over 100% loss-free guarantee (e.g., VoIP, video calls, DNS, gaming)
(B) Financial transaction logging
(C) File download accuracy (e.g., ISO images)
(D) Web page HTML rendering

18. The minimum fixed header length of a standard TCP segment without optional fields is:

(A) 20 Bytes (160 bits)
(B) 8 Bytes
(C) 32 Bytes
(D) 40 Bytes

19. TCP views data transmitted by an application program as:

(A) A continuous, unstructured stream of bytes (Byte-Stream)
(B) Discrete independent message records
(C) Fixed 512-byte blocks
(D) Isolated packet bursts

20. In a TCP segment header, the Sequence Number field identifies:

(A) The byte sequence number of the first data byte in the segment's payload
(B) The total segment count sent so far
(C) The destination process port
(D) The packet priority level

21. The TCP Acknowledgment Number sent by a receiver indicates:

(A) The next expected sequence byte number that the receiver expects to receive (Cumulative ACK)
(B) The sequence number of the byte just received
(C) The receiver's maximum buffer size
(D) The transmission round-trip time

22. Which TCP header flag is set during connection establishment to synchronize sequence numbers?

(A) SYN (Synchronize)
(B) FIN (Finish)
(C) ACK (Acknowledge)
(D) RST (Reset)

23. Which TCP header flag is used to gracefully close or terminate an established TCP connection?

(A) FIN (Finish)
(B) RST (Reset)
(C) PSH (Push)
(D) URG (Urgent)

24. Which TCP header flag is sent to immediately abort and reset an abnormal connection?

(A) RST (Reset)
(B) FIN (Finish)
(C) SYN (Synchronize)
(D) ECE (ECN-Echo)

25. Which TCP header flag tells the receiving system to deliver buffered data to the application immediately without waiting for the buffer to fill?

(A) PSH (Push)
(B) URG (Urgent)
(C) SYN (Synchronize)
(D) ACK (Acknowledge)

26. Connection establishment in TCP is executed using a standard procedure known as:

(A) Three-Way Handshake (SYN -> SYN+ACK -> ACK)
(B) Two-Way Handshake
(C) Four-Way Teardown
(D) Sliding Window Exchange

27. Graceful termination of a full-duplex TCP connection standardly involves how many handshake steps?

(A) Four Steps (FIN -> ACK -> FIN -> ACK)
(B) Three Steps
(C) Two Steps
(D) One Step

28. The 16-bit TCP Window Size field in the segment header is used to implement:

(A) Flow Control (informing sender how many bytes receiver buffer can currently accept)
(B) Congestion Control
(C) Error Detection
(D) Routing metric selection

29. Flow control in TCP protects against:

- **(A)** A fast sender overwhelming and overflowing a slow receiver's internal buffer memory
- **(B)** Routers dropping packets due to intermediate link congestion
- **(C)** Bit errors caused by electrical noise
- **(D)** Man-in-the-middle attacks

30. Congestion control in TCP protects against:

- **(A)** Network routers and intermediate links becoming overloaded and dropping packets
- **(B)** Receiver buffer exhaustion
- **(C)** Host operating system crashes
- **(D)** Unauthorized port scanning

31. The sliding window protocol utilized by TCP operates at the level of:

- **(A)** Individual Bytes
- **(B)** Whole Packets
- **(C)** Frames
- **(D)** Bits

32. In TCP, if a receiver advertises a Receive Window (rwnd) of 0, the sender must:

- **(A)** Halt sending data bytes immediately until an updated window is advertised
- **(B)** Double its transmission rate
- **(C)** Abort the connection with RST
- **(D)** Send full-sized segments anyway

33. To prevent deadlock when an updated non-zero window update packet is lost, the TCP sender runs a:

- **(A)** Persistence Timer (which periodically probes the receiver with 1-byte window probe segments)
- **(B)** Keepalive Timer
- **(C)** Retransmission Timer
- **(D)** TIME_WAIT Timer

34. When a sender transmits a segment, it initializes which timer to guarantee reliability?

- **(A)** Retransmission Timer (RTO - Retransmission Time-Out)
- **(B)** Persistence Timer
- **(C)** TIME_WAIT Timer
- **(D)** Keepalive Timer

35. What happens in TCP when an RTO timer expires before receiving an ACK for a transmitted segment?

- **(A)** The sender retransmits the unacknowledged segment and resets the timer
- **(B)** The connection is terminated immediately
- **(C)** The sender increases its window size
- **(D)** The sender switches to UDP

36. The state that the active closer enters at the end of a TCP teardown before releasing resources is:

- **(A)** TIME_WAIT (usually lasting 2 MSL - Maximum Segment Lifetime)
- **(B)** CLOSE_WAIT
- **(C)** ESTABLISHED
- **(D)** LAST_ACK

37. The primary purpose of the TIME_WAIT state (2 * MSL) is to:

- **(A)** Ensure any delayed or wandering duplicate segments from the old connection dissipate in the network
- **(B)** Speed up immediate port reuse
- **(C)** Download remaining file data
- **(D)** Authenticate user credentials

38. Multiplexing at the transport layer refers to:

- **(A)** Gathering data chunks from multiple application processes, encapsulating them with port headers, and passing them to IP
- **(B)** Splitting an IP packet across multiple Wi-Fi antennas
- **(C)** Merging Ethernet frames onto fiber
- **(D)** Compressing text strings

39. Demultiplexing at the receiving transport layer refers to:

- **(A)** Delivering received segments to the correct application process socket based on destination port numbers
- **(B)** Reordering IP packets
- **(C)** Deciphering TLS passwords
- **(D)** Filtering broadcast packets

40. Stream Control Transmission Protocol (SCTP, RFC 4960) provides which advanced transport capability beyond TCP?

- **(A)** Multi-streaming and Multi-homing
- **(B)** Simpler 4-byte headers
- **(C)** Unreliable multicast only
- **(D)** Hardware layer encryption

41. How does TCP calculate the usable transmission window when both flow control and congestion control are active?

- **(A)** Usable Window = min(rwnd, cwnd), where rwnd is receive window and cwnd is congestion window
- **(B)** Usable Window = rwnd + cwnd
- **(C)** Usable Window = max(rwnd, cwnd)
- **(D)** Usable Window = cwnd * rwnd

42. In TCP congestion control, the Slow Start phase begins with cwnd set to 1 MSS and increases:

- **(A)** Exponentially (doubling cwnd every Round-Trip Time / RTT upon receiving ACKs)
- **(B)** Linearly (by 1 MSS each RTT)
- **(C)** Quadratically
- **(D)** Remains constant

43. When cwnd reaches the Slow Start Threshold (ssthresh), TCP transitions into which phase?

(A) Congestion Avoidance (additive increase: increases cwnd by approximately 1 MSS per RTT)
(B) Fast Recovery
(C) Slow Start Reset
(D) Deadlock State

44. What event triggers the classic TCP Tahoe algorithm to drop cwnd back to 1 MSS?

(A) Detection of packet loss via Retransmission Timeout (RTO)
(B) Arrival of an out-of-order packet
(C) Receipt of a single duplicate ACK
(D) Expiration of the Keepalive timer

45. The Fast Retransmit mechanism allows a TCP sender to detect packet loss before RTO expiry upon receiving:

(A) Three Duplicate ACKs (total of 4 identical ACKs for the same byte)
(B) Two Duplicate ACKs
(C) One Duplicate ACK
(D) Five Duplicate ACKs

46. How does TCP Reno handle packet loss signaled by 3 duplicate ACKs compared to TCP Tahoe?

(A) TCP Reno enters Fast Recovery: halves ssthresh and sets cwnd = ssthresh + 3 MSS, avoiding slow start from 1 MSS
(B) TCP Reno resets cwnd to 1 MSS like Tahoe
(C) TCP Reno increases cwnd to maximum
(D) TCP Reno aborts the connection

47. Why is TCP's congestion control mechanism termed 'Additive Increase / Multiplicative Decrease' (AIMD)?

(A) It gently increases cwnd linearly during normal operation and abruptly cuts cwnd in half upon congestion, guaranteeing stability
(B) It multiplies window by 2 and subtracts 1
(C) It doubles rate constantly
(D) It divides rate by 10

48. In TCP RTT estimation, Jacobson's algorithm computes the Smoothed Round-Trip Time (SRTT) using:

(A) Exponentially Weighted Moving Average: SRTT_new = (1 - alpha) * SRTT_old + alpha * SampleRTT
(B) Simple arithmetic mean of all previous packets
(C) Minimum observed RTT divided by 2
(D) Maximum observed RTT multiplied by 1.5

49. Karn's Algorithm solves the retransmission ambiguity problem in TCP by specifying that:

(A) SampleRTT measurements must never be taken for retransmitted segments when updating RTO estimates
(B) RTO should be reset to zero upon retransmission
(C) All duplicate packets must be ignored
(D) SampleRTT must be doubled for all ACKs

50. What is the primary role of the TCP Selective Acknowledgment (SACK, RFC 2018) option?

(A) It enables the receiver to inform the sender of all non-contiguous byte blocks successfully received, avoiding redundant retransmissions
(B) It allows encrypting packet payloads
(C) It doubles the maximum port capacity
(D) It compresses TCP headers

51. The 'Silly Window Syndrome' occurs when:

(A) Data is exchanged in tiny segment fragments (e.g., 1 byte data with 40 bytes header) due to either sender or receiver inefficiency
(B) The window expands beyond 65535 bytes
(C) A host transmits at gigabit speeds
(D) A router drops all packets

52. Nagle's Algorithm prevents silly window syndrome caused by the sender application by:

(A) Buffering small outgoing data chunks until all outstanding acknowledged data has cleared or a full MSS can be sent
(B) Dropping small packets
(C) Increasing IP packet TTL
(D) Disabling delayed ACKs

53. Clark's Solution prevents silly window syndrome caused by the receiver application by:

(A) Advertising a zero window until the receiver's application buffer has freed at least one full MSS or half the buffer space
(B) Halving the sender's clock rate
(C) Terminating slow client connections
(D) Sending ICMP source quench packets

54. A TCP connection is established between Client port 54321 and Server port 80. Another client from a different IP connects to Server port 80. How does the server differentiate them?

(A) By the unique 4-tuple: (Source IP, Source Port, Destination IP, Destination Port)
(B) By the Destination Port alone
(C) By the MAC address alone
(D) By the TCP Sequence Number

55. What is the Maximum Segment Size (MSS) in TCP?

(A) The largest amount of application payload data that a host can receive in a single TCP segment (excluding IP/TCP headers)
(B) The total size of the Ethernet frame
(C) The maximum window size in bytes
(D) The MTU plus 40 bytes

56. If the network interface MTU is 1500 bytes, what is the default TCP MSS for IPv4 over standard Ethernet?

(A) 1460 Bytes (1500 MTU - 20 bytes IP header - 20 bytes TCP header)
(B) 1500 Bytes
(C) 1480 Bytes
(D) 1440 Bytes

57. Why is the initial sequence number (ISN) in TCP generated pseudo-randomly rather than always starting at 0?

(A) To prevent sequence number prediction attacks (TCP spoofing) and cross-connection packet collisions
(B) To speed up connection establishment
(C) To compress header fields
(D) To synchronize hardware clocks

58. In the TCP 3-way handshake, if Client sends SYN with seq=1000, what will be the Server's SYN+ACK response?

(A) SYN flag set, seq=y (server ISN), and ack=1001 (acknowledging receipt of client's SYN byte)
(B) seq=1000 and ack=1000
(C) seq=1001 and ack=0
(D) seq=0 and ack=1000

59. What happens if a TCP host receives a SYN packet directed to a closed port where no listening process exists?

(A) The host responds with a TCP segment having the RST (Reset) flag set
(B) The host responds with a FIN segment
(C) The host silently ignores the packet
(D) The host sends an ICMP Echo Reply

60. The TCP Window Scale option (RFC 1323) expands the effective maximum receive window from 64 KB up to:

(A) 1 Gigabyte (by shifting the 16-bit window value up to 14 bits left)
(B) 128 Kilobytes
(C) 4 Megabytes
(D) 16 Megabytes

61. A TCP sender with cwnd = 32 KB and RTT = 20 ms has what maximum achievable throughput, assuming no packet loss?

(A) 12.8 Mbps (Throughput = cwnd / RTT = 256,000 bits / 0.02 s = 12.8 Mbps)
(B) 1.6 Mbps
(C) 32 Mbps
(D) 64 Mbps

62. What is the Bandwidth-Delay Product (BDP) of a 1 Gbps transatlantic fiber link with RTT = 100 ms?

(A) 100 Megabits (12.5 Megabytes) of unacknowledged data 'in flight'
(B) 10 Megabits
(C) 1 Gigabit
(D) 50 Megabytes

63. In modern high-speed long-fat networks (LFN), what happens if the TCP window size is smaller than the link's BDP?

(A) The link bandwidth cannot be fully utilized; the sender spends time idling waiting for ACKs
(B) The sender causes massive router buffer overflows
(C) Packets collide and corrupt each other
(D) The TCP connection drops automatically

64. TCP Keepalive timers standardly ping idle connections after how long to ensure the remote peer is still alive?

(A) 2 hours of inactivity (transmitting probe segments every 75 seconds)
(B) 30 seconds
(C) 5 minutes
(D) 24 hours

65. What is the key functional difference between TCP and UDP checksum calculation?

(A) TCP checksum is mandatory in IPv4 and IPv6; UDP checksum is optional in IPv4 but mandatory in IPv6
(B) UDP checksum includes MAC header
(C) TCP checksum uses CRC-32
(D) UDP checksum checks only the header

66. The pseudo-header included in TCP and UDP checksum calculation contains which fields from the IP layer?

(A) Source IP, Destination IP, Protocol/Next-Header number, and Transport Segment Length
(B) Router MAC address and Hop Count
(C) TTL and Type of Service
(D) IP Identification and Flags

67. Why is the IP pseudo-header included in the Transport Layer checksum verification?

(A) To protect against misrouted segments delivered to the wrong host IP address or protocol handler
(B) To encrypt IP addresses
(C) To calculate latency
(D) To optimize switch routing

68. QUIC (Quick UDP Internet Connections, RFC 9000), underlying HTTP/3, operates over which transport protocol?

(A) UDP (providing multiplexed streams without head-of-line blocking and faster 0-RTT handshakes)
(B) TCP
(C) SCTP
(D) Raw IP

69. What primary limitation of TCP does QUIC eliminate when multiplexing multiple streams inside a single connection?

(A) Head-of-Line (HoL) blocking (loss of a single packet halts all streams in TCP, but only affects that single stream in QUIC)
(B) Data corruption
(C) Port limitation
(D) Checksum errors

70. What is the TCP 'Delayed ACK' mechanism designed to achieve?

(A) Reducing network packet overhead by delaying ACK transmission (typically up to 200 ms) so it can piggyback on outgoing data or acknowledge multiple segments
(B) Slowing down greedy servers
(C) Preventing DDoS attacks
(D) Encrypting acknowledgment numbers

71. In TCP connection teardown, if Host A sends FIN and receives ACK from Host B, what state is Host A in?

(A) FIN_WAIT_2 (waiting for Host B to close its application and send its own FIN)
(B) TIME_WAIT
(C) CLOSE_WAIT
(D) CLOSING

72. In Q71, what state does Host B enter upon receiving Host A's FIN and returning an ACK?

(A) CLOSE_WAIT (waiting for its local application to finish sending remaining data and close the socket)
(B) LAST_ACK
(C) TIME_WAIT
(D) ESTABLISHED

73. TCP CUBIC (default in Linux) improves on standard Reno congestion control by scaling the window using:

(A) A cubic function of the elapsed time since the last congestion event, making window growth independent of RTT
(B) An exponential doubling curve
(C) A strict constant step function
(D) A random backoff curve

74. Google's BBR (Bottleneck Bandwidth and RTT) congestion control algorithm differs from loss-based TCP algorithms by:

(A) Measuring actual bottleneck bandwidth and minimum RTT to operate at optimum queue levels without relying on packet loss
(B) Intentionally filling router buffers until packets drop
(C) Using UDP tunneling exclusively
(D) Ignoring round-trip delays

75. Explicit Congestion Notification (ECN, RFC 3168) allows routers to notify TCP endpoints of impending congestion by:

(A) Marking bits in the IP header (ECT and CE bits) instead of dropping packets
(B) Sending ICMP Destination Unreachable packets
(C) Injecting RST packets
(D) Modifying port numbers in transit

76. A TCP sender uses a current cwnd of 16 KB and ssthresh of 32 KB. Packet loss occurs via RTO timeout. Under TCP Tahoe, what are the new values of ssthresh and cwnd (assume MSS = 1 KB)?

(A) ssthresh = 8 KB (half of current cwnd), cwnd = 1 KB (1 MSS)
(B) ssthresh = 16 KB, cwnd = 8 KB
(C) ssthresh = 8 KB, cwnd = 8 KB
(D) ssthresh = 32 KB, cwnd = 16 KB

77. Under TCP Reno, for the same scenario in Q76, if the packet loss is detected by 3 DUPLICATE ACKs instead of RTO, what are the new ssthresh and cwnd values?

(A) ssthresh = 8 KB, cwnd = 11 KB (ssthresh + 3 MSS in Fast Recovery)
(B) ssthresh = 8 KB, cwnd = 1 KB
(C) ssthresh = 16 KB, cwnd = 16 KB
(D) ssthresh = 4 KB, cwnd = 4 KB

78. A TCP connection is transferring a large file. The sender has sent bytes 1001 to 5000. The receiver receives bytes 1001 to 2000 and bytes 3001 to 5000, but bytes 2001 to 3000 are lost. What ACK number does the receiver send back upon receiving byte 5000?

(A) ACK 2001 (cumulative ACK indicating byte 2001 is missing and expected next)
(B) ACK 3001
(C) ACK 5001
(D) ACK 1001

79. In Q78, if the TCP connection has the Selective Acknowledgment (SACK) option enabled, what does the SACK block in the TCP header indicate?

(A) SACK Block: Left Edge = 3001, Right Edge = 5001
(B) SACK Block: Left Edge = 1001, Right Edge = 2001
(C) SACK Block: Left Edge = 2001, Right Edge = 3000
(D) SACK Block: Left Edge = 0, Right Edge = 5000

80. A sender sends segments with Sequence Numbers: Seg 1 (seq=1, length=1000), Seg 2 (seq=1001, length=1000), Seg 3 (seq=2001, length=1000). If Seg 2 is dropped by a router, what are the ACK numbers generated upon receiving Seg 1 and Seg 3?

(A) Seg 1 triggers ACK 1001; Seg 3 triggers duplicate ACK 1001
(B) Seg 1 triggers ACK 1000; Seg 3 triggers ACK 2000
(C) Seg 1 triggers ACK 1001; Seg 3 triggers ACK 3001
(D) Both trigger ACK 1

81. If an estimated RTT is currently 100 ms and a new SampleRTT arrives as 140 ms, using Jacobson's formula with alpha = 0.125, what is the updated SRTT?

(A) SRTT = (1 - 0.125) * 100 + 0.125 * 140 = 87.5 + 17.5 = 105 ms
(B) 120 ms
(C) 140 ms
(D) 100 ms

82. A TCP connection has an RTT of 50 ms and MSS of 1460 bytes. If the receiver advertises rwnd = 64 KB and intermediate routers have infinite buffer capacity, what is the maximum throughput limited solely by rwnd?

(A) Throughput = (65,536 bytes * 8 bits/byte) / 0.050 s = 10.485 Mbps
(B) 1.04 Mbps
(C) 52.4 Mbps
(D) 104.8 Mbps

83. A TCP sender experiences exponential backoff when consecutive retransmission timeouts occur. If the initial RTO is 1 second, what will be the RTO after 3 consecutive retransmissions of the same unacknowledged segment?

(A) 8 seconds (1 s -> 2 s -> 4 s -> 8 s, doubling each timeout up to 64 s)
(B) 3 seconds
(C) 4 seconds
(D) 16 seconds

84. In a SYN flood Denial-of-Service (DoS) attack, an attacker exhausts server resources by transmitting thousands of spoofed SYN packets without sending final ACKs. What defense encodes connection state into the server's initial sequence number?

(A) SYN Cookies (RFC 4987)
(B) Reverse DNS lookup
(C) Dynamic port forwarding
(D) MAC filtering

85. How does SYN Cookie technology prevent connection table exhaustion on a target server?

(A) The server does not allocate memory in its half-open connection table; instead, it generates a cryptographic hash as its ISN and validates it only when the client's final ACK returns
(B) It blocks all incoming SYN packets
(C) It redirects connections to a honeypot
(D) It drops all packets from port 80

86. In TCP connection teardown, if both endpoints initiate the close simultaneously by sending FIN packets at the same time, what sequence of states occurs?

(A) Simultaneous Close: Both enter FIN_WAIT_1 -> both receive FIN and transition to CLOSING -> both receive ACK and transition to TIME_WAIT
(B) Both enter CLOSED immediately
(C) One resets while the other hangs
(D) Deadlock ensues requiring reboot

87. What is the consequence if the TIME_WAIT duration (2 * MSL) is configured to 0 or bypassed using the SO_REUSEADDR socket option unsafely?

(A) A delayed duplicate packet from an old connection could be accepted by a newly opened incarnation on the same socket pair, corrupting data
(B) CPU utilization spikes to 100%
(C) The network interface driver crashes
(D) Subnet routing tables are corrupted

88. Under the TCP Vegas congestion control algorithm, congestion is detected by:

(A) Measuring changes in the observed round-trip throughput and comparing actual throughput against expected throughput, before packet loss occurs
(B) Counting consecutive duplicate ACKs
(C) Waiting for RTO timeouts
(D) Detecting bit errors in payload

89. In an environment with high wireless packet loss (e.g., bit errors on Wi-Fi/LTE), why does standard TCP underperform severely?

(A) Standard TCP mistakenly treats wireless bit-error packet loss as network congestion, inappropriately cutting its cwnd and slashing throughput
(B) TCP cannot run over 802.11 frames
(C) Wi-Fi does not support port addressing
(D) Checksum verification fails on wireless

90. How does TCP handle a segment that arrives with an invalid or unexpected sequence number?

(A) It discards the data and sends an ACK acknowledging the highest in-order contiguous byte received to resynchronize the sender
(B) It resets the connection immediately
(C) It buffers the data permanently
(D) It shuts down the operating system port

91. In the TCP sliding window, what does the 'Usable Window' represent at any given moment?

(A) The number of bytes the sender is permitted to transmit immediately beyond the highest byte already sent (Window Size - Bytes in Flight)
(B) The total size of the sender's RAM buffer
(C) The receiver's maximum packet size
(D) The total bytes acknowledged so far

92. If a sender's window size is 10,000 bytes, LastByteSent = 15,000, and LastByteAcked = 8,000, how many additional bytes is the sender currently allowed to transmit?

(A) 3,000 bytes (Bytes in Flight = 15000 - 8000 = 7000; Usable Window = 10000 - 7000 = 3000)
(B) 2,000 bytes
(C) 5,000 bytes
(D) 0 bytes

93. What security vulnerability arises when an attacker correctly guesses the 32-bit TCP Sequence Number of an active established connection?

(A) TCP Connection Hijacking / Segment Injection (attacker can inject malicious payloads or issue a forged RST to terminate the session)
(B) MAC address spoofing
(C) DNS cache poisoning
(D) Buffer overflow in physical switch

94. What is the effect of setting the TCP_NODELAY socket option in network socket programming?

(A) It disables Nagle's algorithm, forcing segments to be transmitted immediately across the wire without waiting to fill a full MSS or receive pending ACKs
(B) It disables TCP checksum validation
(C) It enables maximum compression
(D) It bypasses the IP routing table

95. Which type of applications specifically benefit from enabling the TCP_NODELAY option?

(A) Interactive, low-latency applications such as online multiplayer games, financial trading terminals, and telnet/SSH keystroke streams
(B) Bulk file transfer servers (FTP)
(C) Video backup archives
(D) BitTorrent downloading

96. In TCP Fast Open (TFO, RFC 7413), how is data transfer accelerated during connection establishment?

(A) The client caches a cryptographic cookie from a previous session and includes application payload directly inside the initial SYN packet
(B) By skipping the DNS lookup step
(C) By removing IP headers
(D) By transmitting over UDP

97. What is the primary function of the Urgent Pointer (URG) in the TCP header?

(A) It points to the sequence number of the byte following out-of-band urgent data that requires immediate priority processing by the receiving application
(B) It marks the highest priority QoS tag
(C) It forces immediate packet retransmission
(D) It indicates an encrypted password payload

98. A link has a capacity of 100 Mbps and RTT of 40 ms. What is the minimum congestion window (cwnd) in bytes required to fully saturate this link?

(A) 500,000 bytes (BDP = 100 * 10^6 * 0.040 / 8 = 500,000 bytes or ~342 MSS of 1460 bytes)
(B) 50,000 bytes
(C) 1,250,000 bytes
(D) 250,000 bytes

99. What happens during a TCP Half-Close scenario where Host A executes shutdown(SHUT_WR)?

(A) Host A sends FIN indicating it has no more data to send, but its receiving socket remains open to accept incoming data from Host B

(B) Both sending and receiving sockets are abruptly closed

(C) Host B is forced to disconnect immediately

(D) All buffered data is discarded immediately

100. In multipath data center networks, Multipath TCP (MPTCP, RFC 8684) allows an application to:

(A) Strip and pool bandwidth simultaneously across multiple network interfaces (e.g., Wi-Fi and 5G cellular) using multiple TCP subflows beneath a single socket

(B) Run TCP over non-IP networks

(C) Bypass TCP flow control rules

(D) Replace Ethernet switches with hubs

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Define the primary functions of the Transport Layer and contrast process-to-process delivery with host-to-host delivery.

Q2. Explain the structure and categorization of Port Numbers in computer networking, detailing Well-Known, Registered, and Ephemeral ports.

Q3. Compare and contrast Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) across all fundamental operating parameters.

Q4. Analyze the complete 8-byte UDP header format and describe the purpose and calculation of each field, including the pseudo-header.

Q5. Draw and describe the complete TCP segment header format, explaining the function of each 32-bit word and control flag.

Q6. Explain the TCP Three-Way Handshake connection establishment process in full technical detail, including sequence number synchronization.

Q7. Describe the four-way handshake procedure for graceful connection termination in TCP, detailing state transitions at both endpoints.

Q8. Explain the purpose, duration, and necessity of the TCP TIME_WAIT state, and analyze the risks of setting 2MSL to zero.

Q9. Explain TCP Sliding Window Flow Control, distinguishing between the Sender Window, Receiver Window (rwnd), and Window Updates.

Q10. Discuss Silly Window Syndrome (SWS), its root causes on sender and receiver sides, and the solutions implemented by Nagle and Clark.

Q11. Explain TCP Congestion Control, detailing the roles of the Congestion Window (cwnd), Slow Start Threshold (ssthresh), and Slow Start phase.

Q12. Describe the Congestion Avoidance, Fast Retransmit, and Fast Recovery phases in TCP Reno.

Q13. Compare TCP Tahoe, TCP Reno, TCP NewReno, and TCP CUBIC congestion control algorithms.

Q14. Explain how TCP calculates and dynamically updates the Retransmission Timeout (RTO) using Jacobson's Algorithm.

Q15. Analyze the Bandwidth-Delay Product (BDP) in high-speed networks and explain why TCP Window Scaling (RFC 1323) is essential.

Q16. Explain Selective Acknowledgment (SACK, RFC 2018) and illustrate how it outperforms standard cumulative ACK when multiple packets are lost.

Q17. Describe the SYN Flood attack mechanism, analyze its impact on server connection tables, and explain the SYN Cookie mitigation technique.

Q18. Explain the functionality of TCP Timers: Retransmission Timer, Persistence Timer, Keepalive Timer, and TIME_WAIT Timer.

Q19. Describe the Stream Control Transmission Protocol (SCTP) and compare its features with TCP and UDP.

Q20. Explain QUIC protocol architecture, highlighting why it is built on UDP and how it solves TCP head-of-line blocking.

Q21. Calculate the maximum achievable throughput of a TCP connection given: RTT = 40 ms, MSS = 1460 bytes, and packet loss rate p = 0.01%, using Mathis's Formula.

Q22. Explain the operation and purpose of Explicit Congestion Notification (ECN, RFC 3168).

Q23. Describe the mechanics of TCP Reset (RST) generation and its legitimate vs malicious uses.

Q24. Explain TCP Maximum Segment Size (MSS) negotiation and how Path MTU Discovery avoids transport-level fragmentation.

Q25. Discuss TCP Half-Close, detailing socket programming semantics and realistic use cases.

Q26. Explain the problem of Bufferbloat in network routers and how modern queue management (AQM / CoDel) mitigates it.

Q27. Compare the Transport Layer OSI reference model specifications with the practical TCP/IP implementation.

Q28. Detail the sequence of TCP socket API system calls executed on both Client and Server to establish a connection, exchange data, and close.

Q29. Explain TCP Fast Open (TFO, RFC 7413) and its impact on web page loading performance.

Q30. Analyze the mathematical stability of the Additive Increase Multiplicative Decrease (AIMD) algorithm using Chiu-Jain vector diagrams.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

1. Connection Model: TCP is connection-oriented, requiring a formal 3-way handshake before data transfer. UDP is connectionless, transmitting datagrams immediately without session setup.
2. Reliability: TCP provides guaranteed delivery via sequence numbering, positive cumulative ACKs, and retransmission timeouts (RTO). UDP is best-effort and unreliable; dropped packets are not recovered.
3. Data Stream Model: TCP provides a continuous, unstructured byte-stream interface. UDP preserves message boundaries, delivering discrete application datagrams.
4. Header Overhead: TCP headers are 20 to 60 bytes with complex control fields. UDP headers are minimal and fixed at exactly 8 bytes.
5. Flow and Congestion Control: TCP implements dynamic sliding window flow control (rwnd) and multi-phase congestion control (AIMD, Slow Start). UDP provides no flow or congestion control.
6. Speed & Latency: UDP delivers low latency and minimal overhead, ideal for real-time applications (VoIP, DNS, video streaming). TCP ensures 100% integrity, ideal for web (HTTP), email (SMTP), and file transfer (FTP).

>>Solution Question Bank

5. Word 5: Checksum (16 bits, mandatory error checking across pseudo-header, TCP header, and data) and Urgent Pointer (16 bits, offset pointing to priority data when URG is set).
6. Optional Words: Options (0-40 bytes, including MSS, Window Scale, SACK Permitted, and Timestamps) padded to a 32-bit boundary.

[Problem Statement]: Explain the purpose, duration, and necessity of the TCP TIME_WAIT state, and analyze the risks of setting 2MSL to zero.
[Detailed Step-by-Step Answer]:
The TIME_WAIT state is entered by whichever endpoint initiates the active close after sending its final ACK in the teardown handshake. The endpoint must remain in TIME_WAIT for a duration equal to twice the Maximum Segment Lifetime (2 * MSL), typically 1 to 2 minutes.

Necessity of TIME_WAIT:
1. Reliable Termination: If the active closer's final ACK is lost in transit, the passive closer will retransmit its FIN. If the active closer had transitioned directly to CLOSED, it would reply with an RST, causing an abnormal abort rather than clean termination.
2. Draining Wandering Duplicates: In IP networks, delayed or re-routed packets from an old connection can arrive long after transmission. 2MSL ensures all delayed duplicate segments completely dissipate from the network, preventing them from corrupting a new connection reincarnated on the same IP-port socket pair.

Risks of Bypassing TIME_WAIT: Setting TIME_WAIT to zero or reusing sockets prematurely risks data corruption when wandering packets from old connections inject erroneous bytes into active sessions.

Solutions:
1. Clark's Solution (Receiver): The receiver is forbidden from advertising tiny window increases. It advertises rwnd = 0 until it can offer a window of at least one full MSS or half of its total buffer space.
2. Nagle's Algorithm (Sender): If unacknowledged data exists in flight, the sender buffers outgoing application data until either a full MSS is accumulated or an ACK for outstanding data is received.

>>**Solution to Question 14:** TCP Congestion Avoidance Algorithms (e.g., TCP Reno)

[Problem Statement]: Compare TCP Tahoe, TCP Reno, TCP NewReno, and TCP CUBIC congestion control algorithms.
[Detailed Step-by-Step Answer]:
Different TCP variants employ distinct strategies to handle congestion and maximize throughput:

1. TCP Tahoe (1988): On any loss (whether by RTO or 3 duplicate ACKs), Tahoe sets ssthresh = cwnd / 2, resets cwnd = 1 MSS, and re-enters Slow Start. This severe penalty creates a 'sawtooth' pattern with prolonged idle periods.
2. TCP Reno (1990): Adds Fast Retransmit and Fast Recovery. On 3 duplicate ACKs, it halves cwnd and stays in Congestion Avoidance, avoiding Slow Start from 1 MSS.
3. TCP NewReno (1999): Fixes Reno's deficiency when multiple packets are dropped from a single window. It remains in Fast Recovery until all segments outstanding at the onset of loss are acknowledged, avoiding premature re-entry to slow start.
4. TCP CUBIC (Modern Linux Default): Replaces linear window growth with a cubic function based on elapsed time since the last congestion event. CUBIC rapidly ramps up window size in high-speed, long-delay networks (LFN) and is less dependent on short RTTs.

>>Problem Statement: Explain how TCP Window Scaling (RFC 1323) is essential.

Max Throughput = 65,535 * 8 / 0.1 s = 5.24 Mbps (utilizing less than 0.1% of link capacity!).

Window Scale Option (RFC 1323):
During handshake, endpoints negotiate a scale factor (0 to 14). Window values are left-shifted by this factor, expanding the maximum usable window up to 1 Gigabyte (65,535 * 2^14), allowing complete saturation of modern ultra-high-speed networks.

[Detailed Step-by-Step Answer]:
TCP relies on four core timers to maintain protocol integrity across asynchronous networks:

1. Retransmission Timer (RTO): Set whenever a segment is sent. If unacknowledged when RTO expires, the segment is retransmitted, RTO doubles (exponential backoff), and cwnd drops to 1 MSS.
2. Persistence Timer: Solves zero-window deadlock. When the receiver advertises rwnd = 0, the sender starts the persistence timer. Upon expiry, the sender transmits a 1-byte Window Probe to prompt an updated window advertisement.
3. Keepalive Timer: Prevents resources from being tied up by orphaned idle connections (e.g., if a client powers off without teardown). After 2 hours of inactivity, the server transmits probe packets. If no response after several probes, the server resets the session.
4. TIME_WAIT Timer (2MSL): Maintains active-close socket resources for twice the maximum segment lifetime to ensure old packets dissipate and final ACKs arrive.

>> Solution to Question 20: Stream Control Transmission Protocol (SCTP) and its features over TCP and UDP. **Solution to Question 20:** The Stream Control Transmission Protocol (SCTP) and UDP and its features over TCP and UDP. blocking.

Additional Advantages: Integrated TLS 1.3 encryption (0-RTT handshakes) and Connection IDs that enable seamless session migration across Wi-Fi and 5G cellular switches without reconnecting.

>>Solution to Question 23: MSS = 1460 bytes, and packet loss rate p = 0

state:

Legitimate Triggers:
1. Non-Listening Port: If a SYN arrives at a destination IP and port where no listening socket process is registered, the OS returns RST+ACK.
2. Half-Open / Orphaned Connections: If Host A crashes and reboots, any incoming segment from an existing session on Host B triggers an RST from Host A because the state was lost.
3. Hard Abort: An application calling close() with SO_LINGER set to timeout=0 discards remaining buffer bytes and sends RST to instantly terminate the session.

Malicious Triggers (TCP Reset Attack):
An eavesdropping attacker who sniffs sequence numbers can forge an RST segment matching the active 4-tuple and in-window sequence number, causing routers or endpoints to tear down legitimate sessions (used in state censorship and session hijacking).

>>Solution Question Bank: TCP Maximum Segment Size (MSS) negotiation and how Path MTU Discovery avoids transport-level fragmentation.

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4. accept(): Blocks until an incoming connection completes the 3-way handshake; returns a new connected socket descriptor.
5. read() / recv() & write() / send(): Exchanges payload bytes with the client.
6. close(): Closes the connected socket descriptor.

Client Side:
1. socket(): Creates an endpoint socket descriptor.
2. connect(): Initiates 3-way handshake to the server's IP and port; blocks until established.
3. write() / send() & read() / recv(): Transmits requests and receives responses.
4. close(): Initiates graceful 4-way teardown handshake.

>>Solution Question 30 Fast Quantum Fourier Transform (FQFT) Additive Increase Multiplicative Decrease (AIMD) algorithm using Chiu-Jain vector diagrams.
